

# **E-Commerce Sales & Customer Analytics Project**

End-to-End Data Analytics Project Using Python, SQL  
& Power BI

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**Tools Used:**

Python | Power BI | SQL

## Table of Contents

| Sl.No | Topic               | Page no |
|-------|---------------------|---------|
| 1.    | Project Overview    | 3       |
| 2.    | Dataset Information | 4       |
| 3.    | Data Cleaning       | 4       |
| 4.    | EDA                 | 10      |
| 5.    | SQL Analysis        | 14      |
| 6.    | Power BI Dashboard  | 17      |
| 7.    | Conclusion          | 19      |

## Project Overview

This project focuses on analysing e-commerce sales data to uncover meaningful business insights related to sales performance, profitability, customer behaviour, product performance, and regional trends.

The project was developed using Python for data cleaning and exploratory data analysis, SQL for business query analysis, and Power BI for interactive dashboard development.

The main objective of this project is to transform raw business data into actionable insights that support business decision-making through data analytics and visualization.

## Project Objectives

- Analyse sales and profit performance across different regions and categories.
- Identify top-performing products and customers.
- Understand monthly sales trends and customer segments.
- Examine the impact of discounts on profitability.
- Build interactive dashboards for executive-level business reporting.
- Demonstrate end-to-end data analytics workflow using Python, SQL, and Power BI.

## Tools & Technologies Used

| Tools/Technology     | Purpose                              |
|----------------------|--------------------------------------|
| Python               | Data Cleaning & exploratory analysis |
| Pandas               | Data manipulation                    |
| Matplotlib & Seaborn | Data Visualization                   |
| SQL                  | Business query analysis              |
| Power BI             | Interactive dashboard development    |

## Dataset Information

The dataset used in this project is the Superstore Sales Dataset, which contains transactional sales data of an e-commerce business.

The dataset includes information related to:

- Orders
- Customers
- Products
- Sales
- Profit
- Discounts
- Regions
- Shipping details

The dataset was used to perform data cleaning, exploratory analysis, SQL business analysis, and interactive dashboard development.

## Data Cleaning & Preprocessing

Data cleaning and preprocessing were performed using Python and Pandas to ensure the dataset was accurate, consistent, and ready for analysis.

Several preprocessing techniques were applied to handle data formatting issues, verify data quality, and prepare the dataset for exploratory data analysis and dashboard development.

## Data Cleaning Steps

- Imported the dataset using Pandas.
- Checked dataset structure using `head()`, `info()`, `shape()`, and `describe()` functions.
- Verified missing values using `isnull().sum()`.
- Checked duplicate records using `duplicated()`.
- Converted date columns into proper datetime format.
- Extracted Month and Year columns for trend analysis.

- Created KPI metrics for Sales, Profit, Orders, and Customers.
- Verified data types for numerical and categorical columns.

## Python Libraries Used

| Library    | Purpose                   |
|------------|---------------------------|
| Pandas     | Data Manipulation         |
| NumPy      | Numerical Operations      |
| Matplotlib | Data Visualization        |
| Seaborn    | Statistical Visualization |

## Importing Libraries & Loading Dataset

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
df=pd.read_csv(r"C:\Users\ADMIN\OneDrive\Desktop\Data Analytics Project\E-Commerce Customer & Sales Analytics Project\Dataset\Sample - Superstore.csv",e
```

*Figure 1: Importing required Python libraries and loading the dataset using Pandas.*

# Dataset Preview

`df.head()`

|   | Row ID | Order ID       | Order Date | Ship Date  | Ship Mode      | Customer ID | Customer Name   | Segment   | Country       | City            | ... | Postal Code | Region | Product ID      | Category        | Sub-Category | Product Name                                      |
|---|--------|----------------|------------|------------|----------------|-------------|-----------------|-----------|---------------|-----------------|-----|-------------|--------|-----------------|-----------------|--------------|---------------------------------------------------|
| 0 | 1      | CA-2016-152156 | 11-08-2016 | 11-11-2016 | Second Class   | CG-12520    | Claire Gute     | Consumer  | United States | Henderson       | ... | 42420       | South  | FUR-BO-10001798 | Furniture       | Bookcases    | Bush Somerset Collection Bookcase                 |
| 1 | 2      | CA-2016-152156 | 11-08-2016 | 11-11-2016 | Second Class   | CG-12520    | Claire Gute     | Consumer  | United States | Henderson       | ... | 42420       | South  | FUR-CH-10000454 | Furniture       | Chairs       | Hon Deluxe Fabric Upholstered Stacking Chairs,... |
| 2 | 3      | CA-2016-138688 | 06-12-2016 | 6/16/2016  | Second Class   | DV-13045    | Darrin Van Huff | Corporate | United States | Los Angeles     | ... | 90036       | West   | OFF-LA-10000240 | Office Supplies | Labels       | Self-Adhesive Address Labels for Typewriters b... |
| 3 | 4      | US-2015-108966 | 10-11-2015 | 10/18/2015 | Standard Class | SO-20335    | Sean O'Donnell  | Consumer  | United States | Fort Lauderdale | ... | 33311       | South  | FUR-TA-10000577 | Furniture       | Tables       | Bretford CR4500 Series Slim Rectangular Table     |
| 4 | 5      | US-2015-108966 | 10-11-2015 | 10/18/2015 | Standard Class | SO-20335    | Sean O'Donnell  | Consumer  | United States | Fort Lauderdale | ... | 33311       | South  | OFF-ST-10000760 | Office Supplies | Storage      | Eldon Fold 'N Roll Cart                           |

Figure 2: Preview of the Superstore sales dataset using head().

## Dataset Information

```
df.info()
```

```
<class 'pandas.DataFrame'>  
RangeIndex: 9994 entries, 0 to 9993  
Data columns (total 21 columns):  
#   Column                Non-Null Count  Dtype    
---  ---                     -  
0   Row ID                9994 non-null  int64    
1   Order ID              9994 non-null  str       
2   Order Date            9994 non-null  str       
3   Ship Date              9994 non-null  str       
4   Ship Mode              9994 non-null  str       
5   Customer ID            9994 non-null  str       
6   Customer Name          9994 non-null  str       
7   Segment                9994 non-null  str       
8   Country                9994 non-null  str       
9   City                   9994 non-null  str       
10  State                  9994 non-null  str       
11  Postal Code            9994 non-null  int64     
12  Region                 9994 non-null  str       
13  Product ID             9994 non-null  str       
14  Category               9994 non-null  str       
15  Sub-Category           9994 non-null  str       
16  Product Name           9994 non-null  str       
17  Sales                  9994 non-null  float64   
18  Quantity               9994 non-null  int64     
19  Discount               9994 non-null  float64   
20  Profit                 9994 non-null  float64   
dtypes: float64(3), int64(3), str(15)  
memory usage: 1.6 MB
```

*Figure 3: Dataset structure and data types using info().*

## Missing Value Analysis

```
df.isnull().sum()
```

```
Row ID          0
Order ID        0
Order Date      0
Ship Date       0
Ship Mode       0
Customer ID     0
Customer Name   0
Segment        0
Country        0
City           0
State          0
Postal Code     0
Region         0
Product ID     0
Category       0
Sub-Category   0
Product Name   0
Sales          0
Quantity       0
Discount       0
Profit         0
dtype: int64
```

*Figure 4: Verification of missing values in the dataset.*

## Duplicate Record Analysis

```
df.duplicated().sum()
```

```
np.int64(0)
```

*Figure 5: Duplicate record verification in the dataset.*



## Statistical Summary

```
df.describe()
```

|              | Row ID      | Postal Code  | Sales        | Quantity    | Discount    | Profit       |
|--------------|-------------|--------------|--------------|-------------|-------------|--------------|
| <b>count</b> | 9994.000000 | 9994.000000  | 9994.000000  | 9994.000000 | 9994.000000 | 9994.000000  |
| <b>mean</b>  | 4997.500000 | 55190.379428 | 229.858001   | 3.789574    | 0.156203    | 28.656896    |
| <b>std</b>   | 2885.163629 | 32063.693350 | 623.245101   | 2.225110    | 0.206452    | 234.260108   |
| <b>min</b>   | 1.000000    | 1040.000000  | 0.444000     | 1.000000    | 0.000000    | -6599.978000 |
| <b>25%</b>   | 2499.250000 | 23223.000000 | 17.280000    | 2.000000    | 0.000000    | 1.728750     |
| <b>50%</b>   | 4997.500000 | 56430.500000 | 54.490000    | 3.000000    | 0.200000    | 8.666500     |
| <b>75%</b>   | 7495.750000 | 90008.000000 | 209.940000   | 5.000000    | 0.200000    | 29.364000    |
| <b>max</b>   | 9994.000000 | 99301.000000 | 22638.480000 | 14.000000   | 0.800000    | 8399.976000  |

*Figure 6: Statistical summary of numerical columns using describe().*

## Feature Engineering

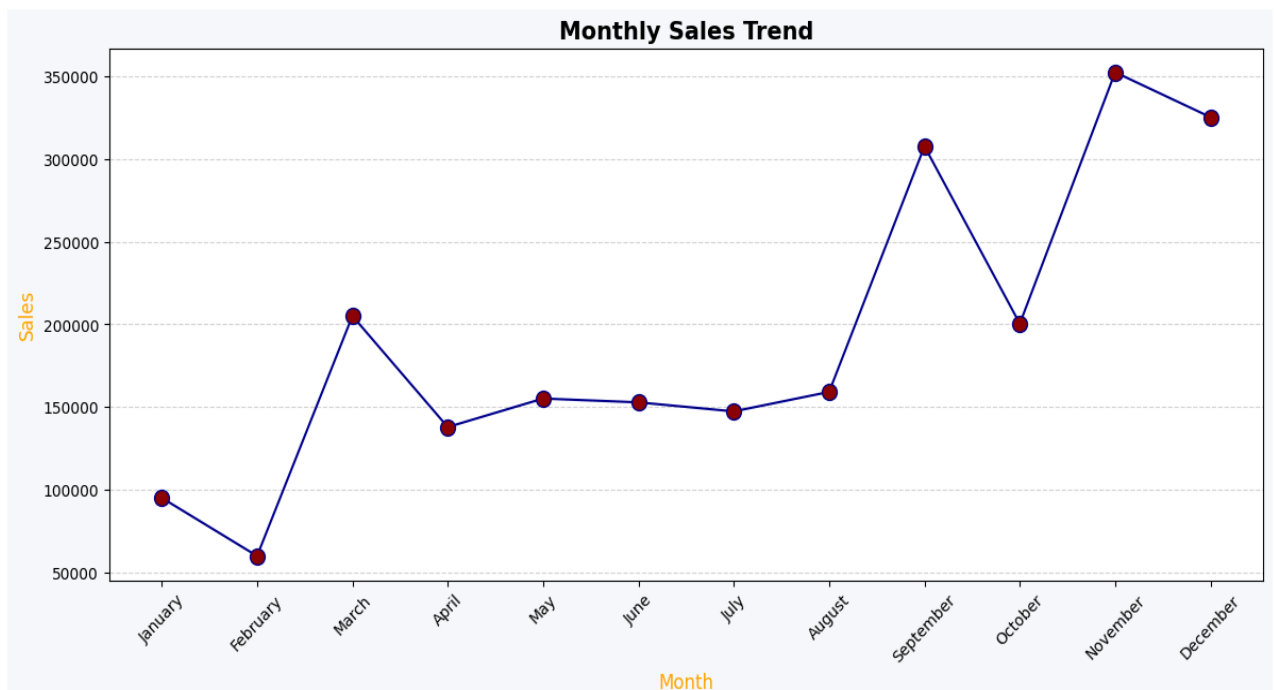
Additional columns such as Month and Year were created from the Order Date column to support time-series and trend analysis.

# Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed using Python to identify sales patterns, profitability trends, customer behaviour, and regional performance.

Various visualizations were created using Matplotlib and Seaborn to better understand the business data and uncover meaningful insights.

## Monthly Sales Trend

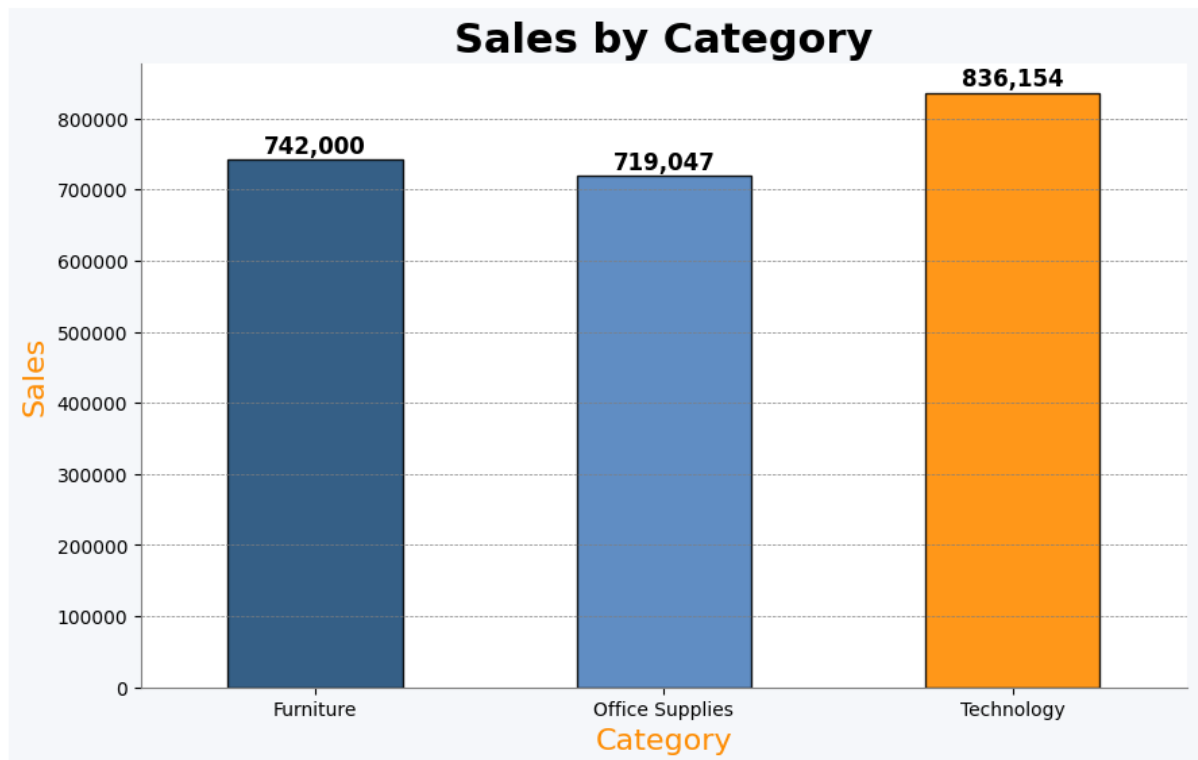


*Figure 7: Monthly sales trend analysis using Python visualization.*

## Insight

The monthly sales trend analysis revealed fluctuations in sales performance throughout the year. Sales increased significantly during September, November, and December, indicating strong year-end business activity and seasonal demand patterns.

## Sales by Category



*Figure 8: Sales distribution across product categories.*

## Insight

The Technology category generated the highest sales revenue compared to Furniture and Office Supplies, highlighting strong customer demand for technology-related products.

# Profit by Region

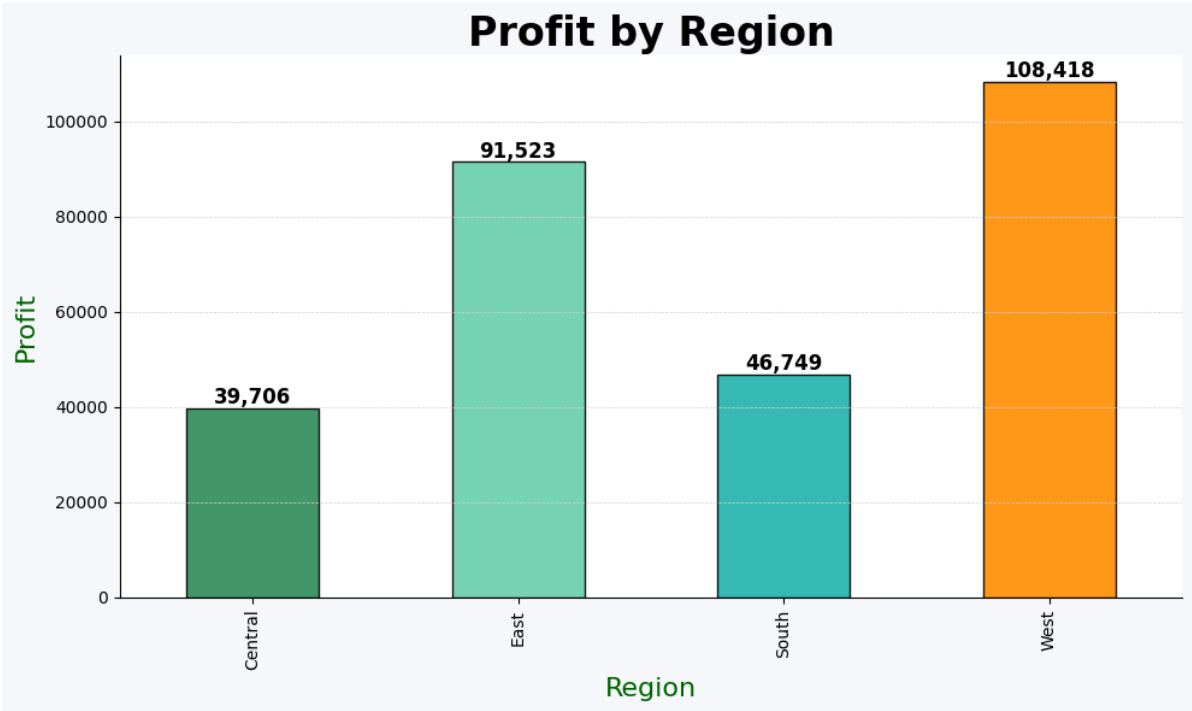
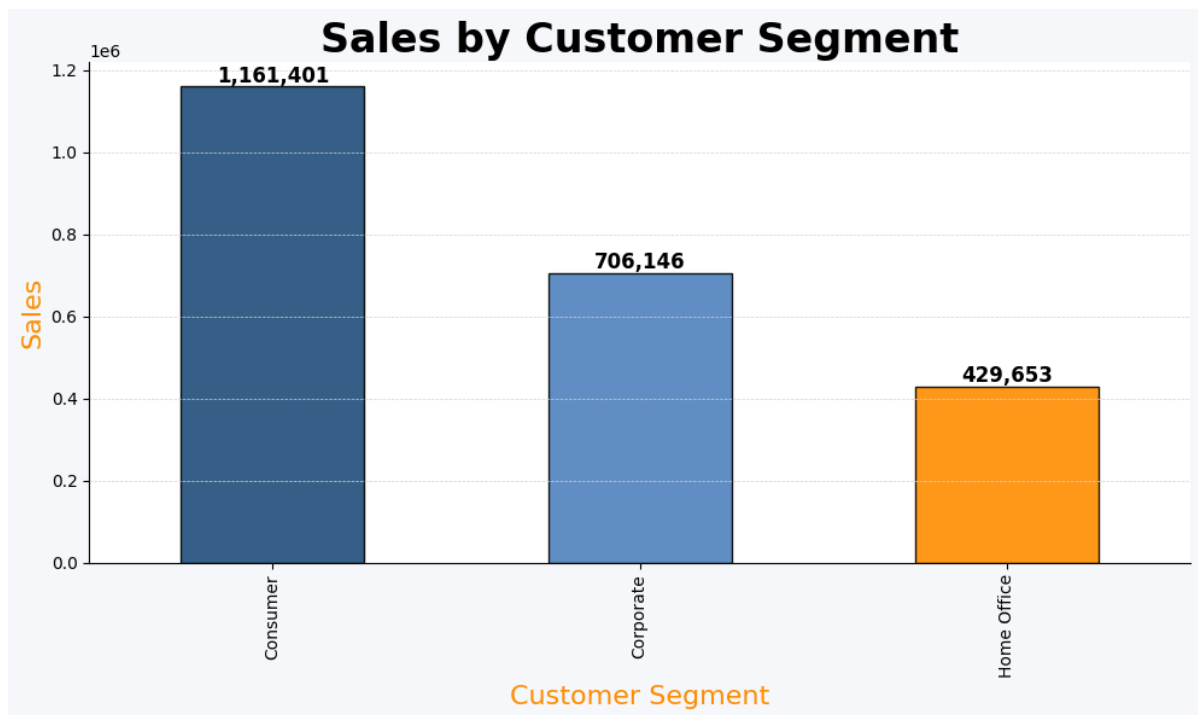


Figure 9: Regional profit analysis.

## Insight

The West region contributed the highest overall profit, while the Central and South regions showed comparatively lower profitability. This indicates stronger business performance in the western region.

## Sales by Segment



*Figure 10: Sales contribution by customer segments.*

## Insight

The Consumer segment contributed the highest sales share among all customer segments, indicating that individual consumers represented the largest portion of business revenue.

## Key Observation

- Technology category achieved the highest sales.
- Consumer segment generated maximum revenue contribution.
- Sales increased significantly during year-end months.
- West region showed the strongest profitability performance.

## SQL Business Analysis

SQL was used to perform business-oriented analysis on the Superstore dataset. Various queries were written to analyse sales performance, profitability, customer behaviour, and regional trends.

The SQL analysis helped extract meaningful insights that support data-driven business decisions.

### Region-wise Profit Analysis

#### Business Question:

Which region generated the highest overall profit?

#### Query:

```
SELECT Region, SUM(Profit) AS Total_Profit FROM  
superstore GROUP BY Region ORDER BY Total_Profit  
DESC;
```

#### Insight:

The West region generated the highest total profit among all regions, indicating stronger business performance and profitability in that region.

### Category-wise Sales Analysis

#### Business Question:

Which product category generated the highest sales?

#### Query:

```
SELECT Category, SUM(Sales) AS Total_Sales FROM  
superstore GROUP BY Category ORDER BY Total_Sales  
DESC;
```

## Insight:

The Technology category generated the highest sales revenue, showing strong customer demand for technology products.

## Customer Segment Analysis

### Business Question:

Which customer segment contributed the highest sales?

### Query:

```
SELECT Segment, SUM(Sales) AS Total_Sales FROM  
superstore GROUP BY Segment ORDER BY Total_Sales  
DESC;
```

## Insight:

The Consumer segment contributed the highest share of total sales, making it the most significant customer segment for the business.

## Monthly Sales Trend

### Business Question:

How did sales perform across different months?

### Query:

```
SELECT MONTHNAME(Order_Date_Formatted) AS  
Month, SUM(Sales) AS Total_Sales FROM superstore GROUP  
BY MONTHNAME(Order_Date_Formatted) ORDER BY  
MONTH(Order_Date_Formatted);
```

## Insight:

Sales showed stronger performance during the later months of the year, especially during November and December, indicating seasonal business growth.

## Discount Impact on Profit

### Business Question:

How do discounts impact profitability?

### Query:

```
SELECT Discount,AVG(Profit) AS Avg_Profit FROM  
superstore GROUP BY Discount ORDER BY Discount;
```

## Insight:

Higher discount levels were associated with lower average profit, indicating that excessive discounts negatively affected profitability.



# Power BI Dashboard

Interactive dashboards were developed using Power BI to visualize sales performance, customer insights, regional profitability, and operational trends.

The dashboards provide an executive-level overview of business performance and allow dynamic filtering using slicers and interactive visuals.

## Executive Overview Dashboard



Figure 11: Executive Overview Dashboard developed in Power BI.

The Executive Overview Dashboard provides high-level business insights related to sales, profit, customer segments, category performance, and regional profitability.

Interactive slicers allow users to dynamically filter dashboard visuals by region, category, segment, and year.

# Detailed Insights Dashboard

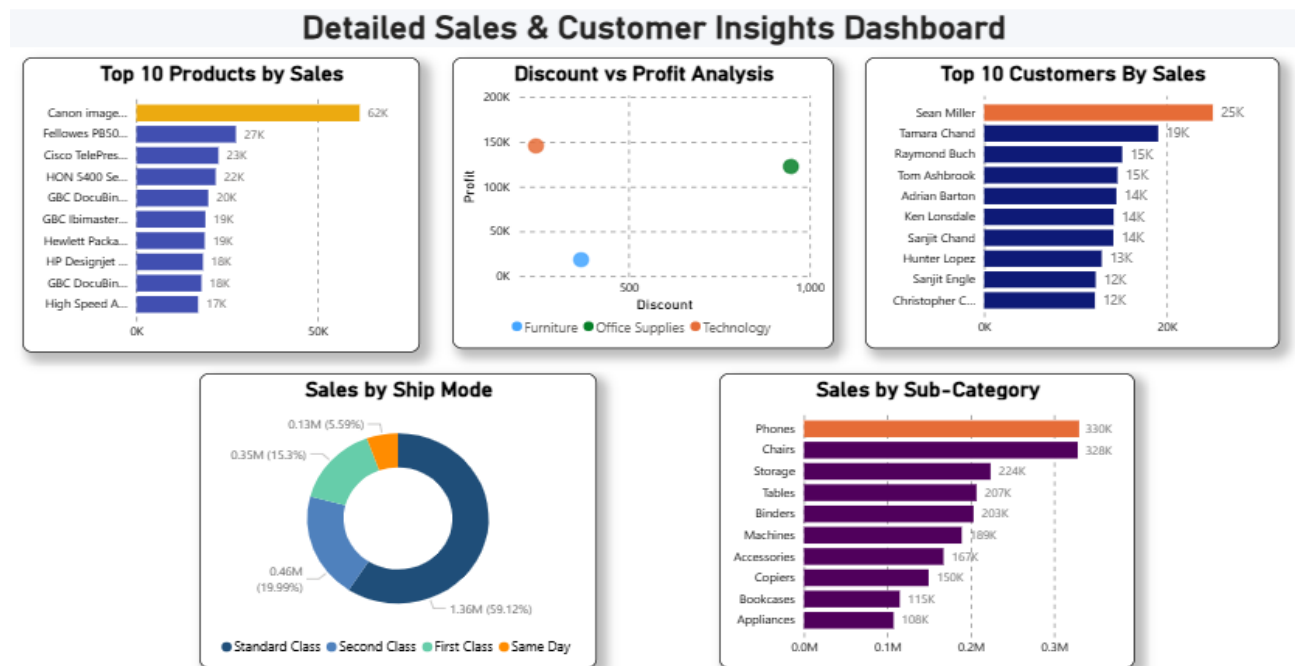


Figure 12: Detailed Insights Dashboard developed in Power BI.

The Detailed Insights Dashboard focuses on deeper business analysis including top-performing products, customer analysis, discount impact on profitability, sub-category performance, and shipping analysis.

The dashboard provides detailed operational and customer-level insights to support business strategy and decision-making.

## Dashboard Features

- KPI cards for Sales, Profit, Orders, and Customers
- Interactive slicers for dynamic filtering
- Trend analysis using line charts
- Geographic sales analysis using map visualization
- Customer and product performance analysis
- Profitability and discount impact analysis

## Key Business Insights

- Technology category generated the highest overall sales revenue.
- The West region contributed the highest profit among all regions.
- Consumer segment represented the largest customer contribution to sales.
- Sales activity increased significantly during the year-end months.
- Higher discount levels negatively impacted profitability.
- California generated the strongest sales performance among all states.
- Standard Class was the most commonly used shipping mode.

## Conclusion

This project successfully demonstrated an end-to-end data analytics workflow using Python, SQL, and Power BI.

The project involved data cleaning, exploratory data analysis, SQL business analysis, and interactive dashboard development to uncover valuable business insights related to sales performance, profitability, customer behaviour, and regional trends.

The dashboards and analytical findings provide meaningful support for data-driven business decision-making and performance evaluation.

## Future Improvements

- Implement predictive sales forecasting using Machine Learning techniques.
- Develop customer segmentation models for targeted marketing analysis.
- Integrate real-time business data into dashboards.
- Create automated reporting and alert systems.
- Expand dashboard functionality with advanced KPI tracking.